b) ~~Describe the concept of Pigeon - hole Principle with example~~

1) Show that $1^2 + 2^2 + 3^3 + \ldots\ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n > 1$

→

Let the statement be denoted by $S(n)$

$n = 1$

$$1^2 = \frac{1(1+1)(2(1)+1)}{6}$$

$$\frac{2(3)}{6} = \frac{6}{6}$$

$$= 1 \quad \text{true}.$$

Assume that $S(k)$ is true.

$$1^3 + 2^3 + 3^2 + \ldots\ldots k^3 = \frac{k^2(k+1)^2}{4}$$

$n + k + 1$

$$\frac{k(k+1+1)(2(k+1)+1)}{6}$$

$$\frac{k(2k^2)(2k+1)+1)}{6}$$

$$\frac{(k+1)^2(k+2)^2}{4}$$

By Principle of mathmatical induction $S(n)$ is true

1 a] Let $U = \{a, b, c, d, e, f\}$ and $A = \{a, b, c\}$ $B = \{b, e, f\}$ Find

i) $B' - A'$

ii) $(A \cap B)'$

⟶

Given $U = \{a, b, c, d, e, f\}$

$A = \{a, b, c\}$

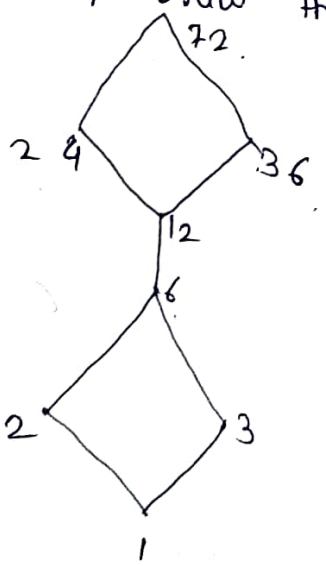
$B = \{b, e, f\}$

$A' = \{d, e, f\}$ $B' = \{a, c, d\}$

$\therefore B' - A' = \{a, c\}$

$\therefore (A \cap B)' = A \cup B = \{a, c, d, e, f\}$ — Ans

b] Let $X = \{1, 2, 3, 6, 12, 24, 36\}$ and relation $\leq$ be such that $x \leq y$ if "x divides y" Draw the Hasse diagram



Write negation and contrapositive of the statement 'If I have time and I am not fired, then I will go to the market,

→

Let p : I have time

q : I am not too fired

r : I will go to store

∴ Give statement is $(p \wedge q) \rightarrow r$

a) Converse $r \rightarrow (p \wedge q)$

If I have go the market then I have time and I am not too fired

b) Inverse $\sim(p \wedge q) \rightarrow r$. $(\sim p \vee \sim q) \rightarrow r$

If I don't have time or I am too fired, then I will not go to the market.

c) Contrapositive. $\sim r \rightarrow \sim(p \wedge q)$ $\sim r \rightarrow (\sim p \vee \sim q)$

If I do not go to the market, then I don't have time or I am not fired

d) Negation $\sim[(p \wedge q) \rightarrow r] = (p \wedge q) \wedge \sim r$

I have time and I am not too fired I will go to the market.

Let F be the set of all one-one and onto mapping from X to X, Where $X = \{1,2,3\}$ Find all stement of F and find inverse of each element.

→

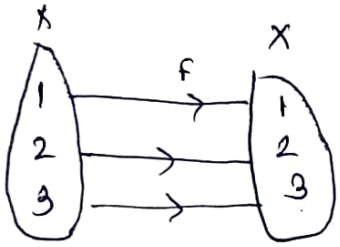
Given $X = \{1,2,3\}$ and $f: X \to X$

∴ X has three element.

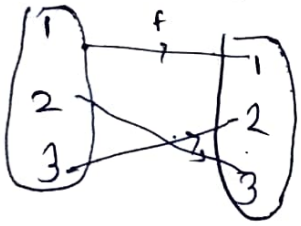
∴ Number of one-one and onto mapping from X to $X = 3$

$3 \times 2 \times 1 = 6$

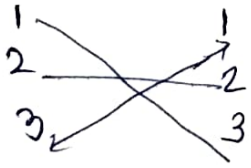
i) $F_1 = \{(1,1)(2,2)(3,3)\}$



ii) $f_2 = \{(1,1)(2,3)(3,2)\}$



iii) $f_3 = \{(1,3)(2,2)(3,1)\}$



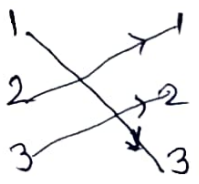
iv) $f_4 = \{(1,2)(2,1)(3,3)\}$



vi) $F_6 = \{(1,3)(2,1)(3,2)\}$



vi) $F_6 = \{(1,3)(2,1)(3,2)\}$



$f = \{f_1, f_2, f_3, f_4, f_5, f_6\}$

Now we will find the inverse of each element

r_1^{-1}

$\therefore r_1^{-1} = r_1 = 1$

v) $r_5^{-1} = r_6$



vi) $r_6^{-1} = r_5$



The quotient set of Z by R is $\frac{Z}{R} = \{[0]_R, [1]_R, [2]_R\}$

Note that $[0]_R = [3]_R = [6]_R = \ldots$

$[1]_R = [4]_R = [7]_R = \ldots$

$\therefore Z = [0]_R \cup [1]_R \cup [2]_R$

We get a partition of Z by partitions $[0]_R, [1]_R, [2]_R$.

4] Let R be the set of integers and R be the relation defined by $R = \{(x, y) : x \in Z, y \in Z \wedge (x-y)$ is divisible by $6\}$.

→

(This relation Z is also called "congruence modulo 6")

This equivalence classes generated by R are

$[0]_R = \{\ldots -6, -3, 0, 3, 6 \ldots\}$

= set equivalence classes generated by R are of element of Z related to 0

$[1]_R = \{\ldots -5, -2, 1, 4, 7 \ldots\}$

set of element of Z related to 1

$[2]_R = \{\ldots -4, -1, 2, 5, 8 \ldots\}$